

Azure Cost Management Budget & Anomaly Alerting

Control cloud spending with budgets, cost analysis, ownership tags, and alerts that highlight unusual spending.

USE CASE 01

Alert department owners when spending approaches or exceeds the approved budget.

USE CASE 02

Identify the top five monthly cost drivers and investigate unusual changes.

STUDENT SCOPE

Use Azure Cost Management with student credits and avoid unnecessary paid resources.

Team Members:

T.Naga Bala Nandini-2400031197

Sk.Kalesha vali-2400031180

A.Kuladeep reddy-2400031192

K.Eswar naga sai-2400031152

Cloud spend can grow before anyone notices

Budget visibility

Without a defined budget and threshold alerts, teams may discover overspending only after the cost has already accumulated.

Unclear ownership

Untagged resources make it difficult to map costs to a department, team, project or owner for chargeback.

Unexpected spikes

A VM, database, storage account or data transfer pattern can create a sudden increase that looks unusual compared with normal usage.

Bottlenecks students may encounter

- Alerts can arrive after spend has already occurred.
- Missing tags make cost allocation incomplete.
- Student credits are limited, so testing must avoid unnecessary resources.

What Azure Cost Management does

- Azure Cost Management helps users understand where cloud money is being spent and provides tools for budgets, cost analysis and alerts.
- Budgets define an expected spending limit and can trigger notifications when configured thresholds are reached.
- Cost analysis helps break spending down by resource, resource group, service, subscription, tag and time period.
- Anomaly monitoring focuses attention on unusual changes so the team can investigate the underlying resource or usage pattern.

INPUT

Usage + resource metadata + subscription cost data

CONTROL

Budget thresholds + ownership tags

OUTPUT

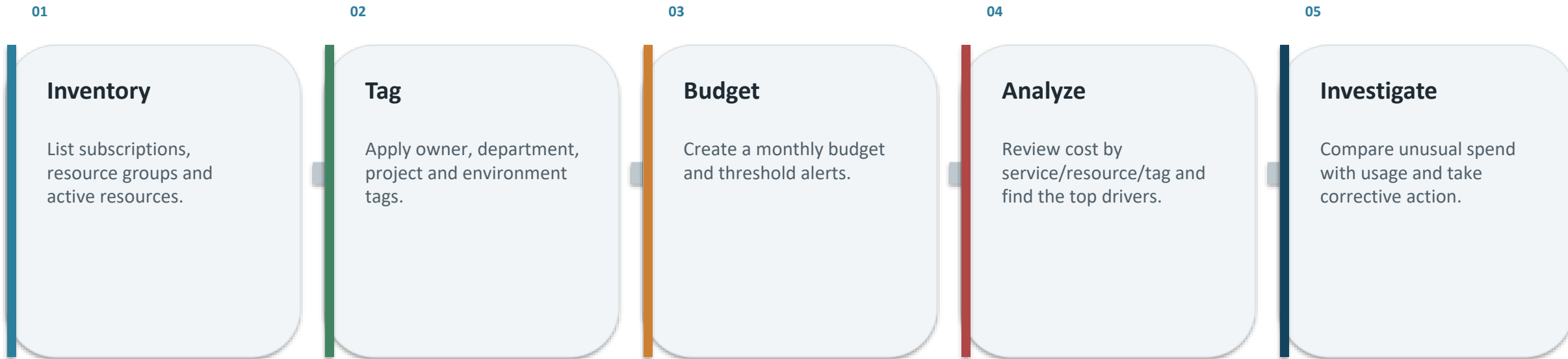
Alerts + monthly cost-driver report

A practical approach to Azure cost visibility

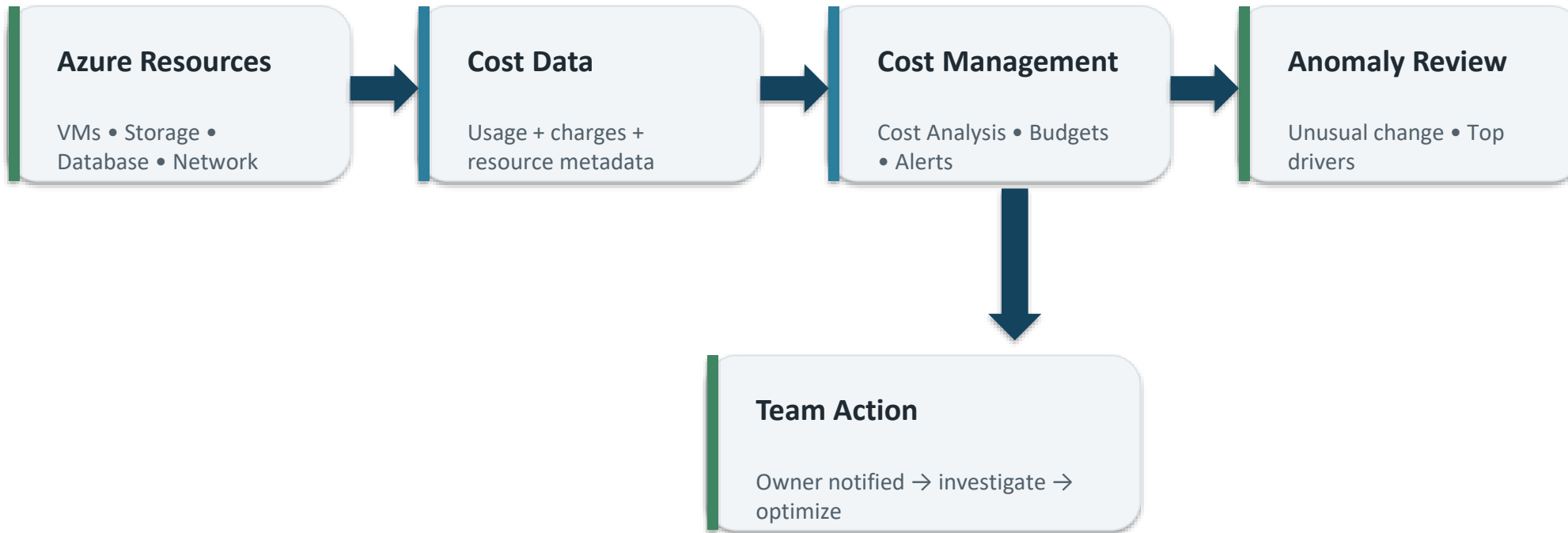
Azure Cost Management Budget and Anomaly Alerting is designed to help student teams monitor cloud expenditure, define spending limits, and identify unusual cost changes. The project combines Cost Analysis, budgets, alerts, and resource tags to connect cloud usage with departments, teams, projects, and owners. Monthly cost analysis is used to identify the top five cost drivers, while anomaly investigation helps explain unexpected spikes. The workflow provides a simple cycle of measuring spend, detecting a signal, investigating the responsible resource, and taking corrective action while keeping student Azure credits in mind.

Measure → Budget → Alert → Investigate → Act

A simple five-stage cost-control method



How the cost monitoring pieces connect



Central idea: measure the spend, assign ownership, set thresholds, detect unusual changes, then act on the evidence.

From cloud usage to a corrective action



What counts as an anomaly?

Baseline

Establish the normal spending pattern for the chosen subscription, resource group, service or resource.

Signal

A sudden increase, unexpected new cost, or abnormal change becomes a candidate for investigation.

Investigation

Check the resource, usage, deployment time, tags and recent configuration changes before deciding what happened.

Example

- Normal monthly compute spend: approximately ₹X.
- Current month shows a sharp increase after a new VM was deployed.
- Cost Analysis identifies the VM as one of the top five cost drivers.
- The owner checks the VM size, running hours and whether it is still required.

Student-friendly Azure Portal setup

01

Open Cost Management + Billing

Choose the correct student subscription.

02

Review Cost Analysis

Set the time range and group costs by service/resource.

03

Add cost allocation tags

Use consistent values such as Department, Team, Project and Environment.

04

Create a budget

Set a monthly amount that fits the available student credits.

05

Configure alerts

Choose threshold percentages and notification recipients.

06

Review anomalies

Use cost trends and unusual changes to identify resources needing investigation.

Identify the top five cost drivers

Rank	Cost driver	What to inspect
1	Example: Virtual Machines	VM size, running hours, idle instances
2	Example: Storage	Capacity growth, transactions, snapshots
3	Example: Database	Tier, compute hours, provisioned capacity
4	Example: Networking	Data transfer and unnecessary traffic
5	Example: Other services	Unexpected deployments or configuration changes

Current implementation: Cost Analysis is configured as “Monthly Cost by Service”. The new student subscription currently has no reported billable cost data, s

LIVE IMPLEMENTATION — ACTUAL AZURE CONFIGURATION

Configuration	Implemented value
Subscription	Azure for Students
Resource group	RG-CostManagement
Storage account	costalertstorage001
Monthly budget	₹10
Warning threshold	80% / ₹8
Critical threshold	100% / ₹10
Cost Analysis view	Monthly Cost by Service
Anomaly rule	Daily anomaly by resource group — Active
Notifications	Email to configured recipients
Cost data status	No billable cost reported yet

LIVE AZURE IMPLEMENTATION EVIDENCE

01 Resource + Tags

RG-CostManagement / costalertstorage001
Department=IT • Project=CostManagement • Owner=Student

02 Budget + Threshold Alerts

₹10 monthly • 80%=₹8 • 100%=₹10
Email notifications configured

03 Cost Analysis

Monthly Cost by Service • September 2026
No billable cost data reported yet

04 Anomaly Detection

Daily anomaly by resource group
Status: Active • Email notification

What the project gives the team

Budget control

A defined spending target with threshold-based notifications.

Cost visibility

A monthly view of where the subscription money is going.

Ownership

Tags make it easier to connect resources with teams and projects.

Anomaly investigation

Unusual changes become review items instead of remaining hidden.

Student constraint

Keep test resources small, short-lived and within the credit limit.

Evidence

Capture portal screenshots of budgets, alerts, tags and cost analysis.

Turning cloud spending into an actionable feedback loop

- Cost Management provides a central view of where Azure spending is occurring.
- Budgets and threshold alerts provide an early signal when spending approaches a planned limit.
- Consistent resource tags improve ownership and make chargeback or team-level analysis easier.
- Monthly Cost Analysis helps the team identify the top five cost drivers and investigate unusual changes.
- For student projects, monitoring is especially important because Azure credits are limited.

Key takeaway: visibility + ownership + alerts enable timely cost-control decisions.

Thank You

Azure Cost Management
Budget & Anomaly Alerting